

```
In [2]: !pip install statsmodels
```

```
Requirement already satisfied: statsmodels in c:\users\use\appdata\local\programs\python\python311\lib\site-packages (0.14.6)
Requirement already satisfied: numpy<3,>=1.22.3 in c:\users\use\appdata\local\programs\python\python311\lib\site-packages (from statsmodels) (1.24.3)
Requirement already satisfied: scipy!=1.9.2,>=1.8 in c:\users\use\appdata\local\programs\python\python311\lib\site-packages (from statsmodels) (1.11.2)
Requirement already satisfied: pandas!=2.1.0,>=1.4 in c:\users\use\appdata\local\programs\python\python311\lib\site-packages (from statsmodels) (2.0.1)
Requirement already satisfied: patsy>=0.5.6 in c:\users\use\appdata\local\programs\python\python311\lib\site-packages (from statsmodels) (1.0.2)
Requirement already satisfied: packaging>=21.3 in c:\users\use\appdata\roaming\python\python311\site-packages (from statsmodels) (23.1)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\use\appdata\local\programs\python\python311\lib\site-packages (from pandas!=2.1.0,>=1.4->statsmodels) (2.8.2)
Requirement already satisfied: pytz>=2020.1 in c:\users\use\appdata\local\programs\python\python311\lib\site-packages (from pandas!=2.1.0,>=1.4->statsmodels) (2023.3)
Requirement already satisfied: tzdata>=2022.1 in c:\users\use\appdata\local\programs\python\python311\lib\site-packages (from pandas!=2.1.0,>=1.4->statsmodels) (2023.3)
Requirement already satisfied: six>=1.5 in c:\users\use\appdata\local\programs\python\python311\lib\site-packages (from python-dateutil>=2.8.2->pandas!=2.1.0,>=1.4->statsmodels) (1.16.0)
[notice] A new release of pip available: 22.3.1 -> 26.1.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.tsa.stattools import adfuller
```

```
In [2]: df = pd.read_csv("data/processed/monthly_housing_train.csv")
df["date"] = pd.to_datetime(df["date"])
df = df.sort_values("date")
df = df.set_index("date")
y = df["hpi"].dropna()

# rolling mean 與 rolling standard deviation
y_rolling_mean_12 = y.rolling(window=12).mean()
y_rolling_std_12 = y.rolling(window=12).std()
```

Stationary inspection

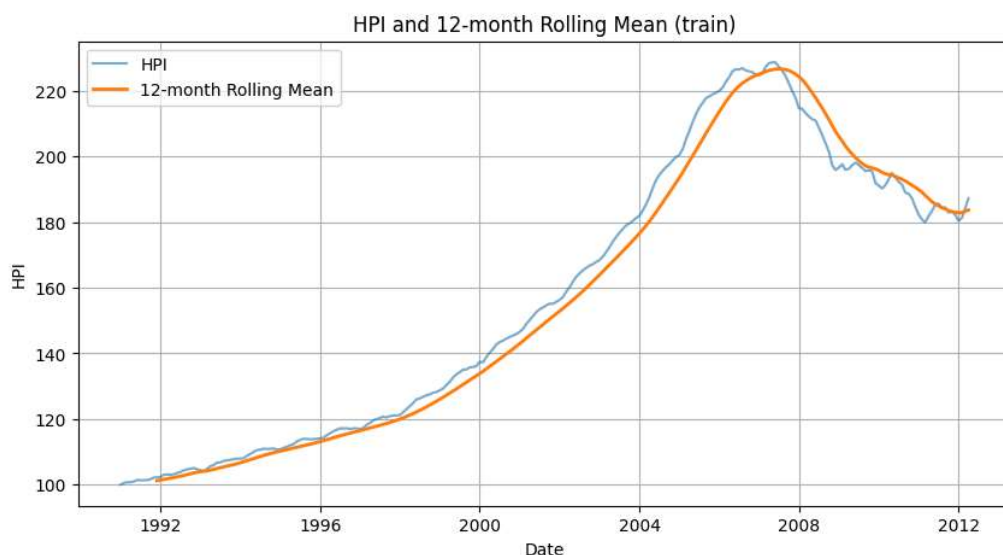
12 month rolling mean

```
In [3]: plt.figure(figsize=(10, 5))

plt.plot(y.index, y, label="HPI", alpha=0.6)
plt.plot(y_rolling_mean_12.index, y_rolling_mean_12,
        label="12-month Rolling Mean", linewidth=2)

plt.title("HPI and 12-month Rolling Mean (train)")
plt.xlabel("Date")
plt.ylabel("HPI")
plt.legend()
plt.grid(True)

plt.show()
```



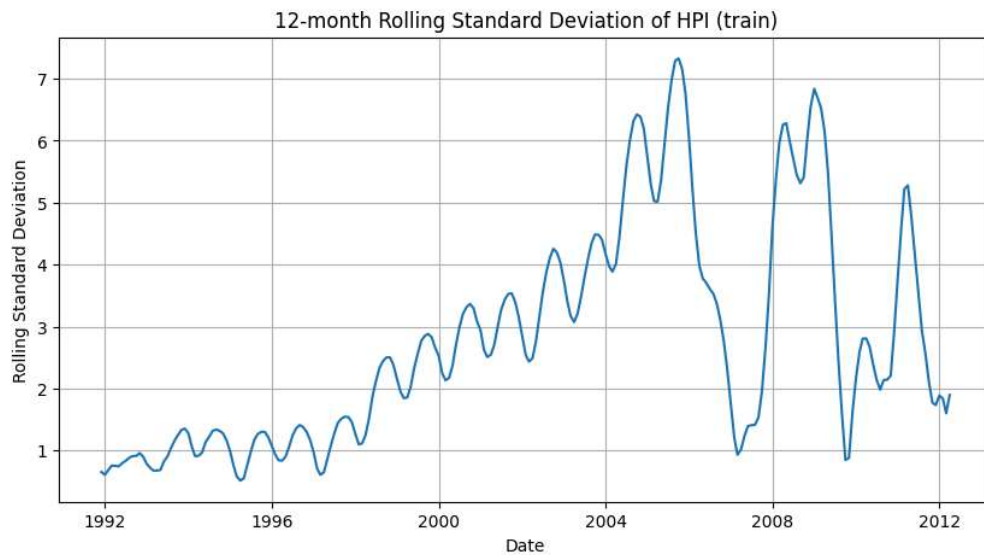
12 month rolling std

```
In [5]: plt.figure(figsize=(10, 5))

plt.plot(y_rolling_std_12.index, y_rolling_std_12)

plt.title("12-month Rolling Standard Deviation of HPI (train)")
plt.xlabel("Date")
plt.ylabel("Rolling Standard Deviation")
plt.grid(True)

plt.show()
```



ACF plot

```
In [6]: max_lag = 48

acf_values = [y.autocorr(lag=lag) for lag in range(max_lag + 1)]

plt.figure(figsize=(10, 5))

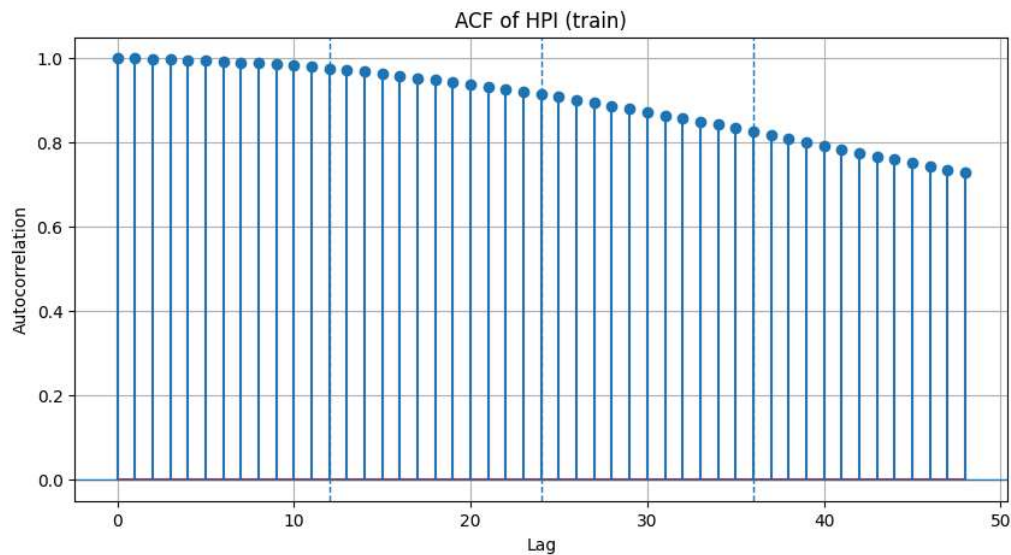
plt.stem(range(max_lag + 1), acf_values)

for lag in [12, 24, 36]:
    plt.axvline(x=lag, linestyle="--", linewidth=1)

plt.axhline(y=0, linewidth=1)

plt.title("ACF of HPI (train)")
plt.xlabel("Lag")
plt.ylabel("Autocorrelation")
plt.grid(True)

plt.show()
```



monthly seasonal plot

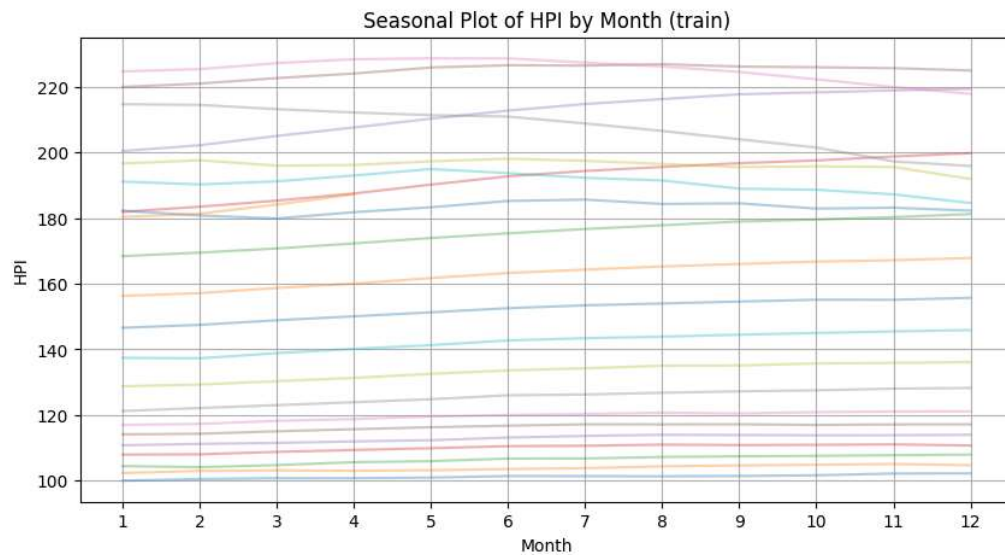
```
In [7]: season_df = y.to_frame(name="hpi")
season_df["year"] = season_df.index.year
season_df["month"] = season_df.index.month

plt.figure(figsize=(10, 5))

for year, group in season_df.groupby("year"):
    plt.plot(group["month"], group["hpi"], alpha=0.35)

plt.title("Seasonal Plot of HPI by Month (train)")
plt.xlabel("Month")
plt.ylabel("HPI")
plt.xticks(range(1, 13))
plt.grid(True)

plt.show()
```



```
In [8]: # ADF test
for reg in ["c", "ct"]:
    result = adfuller(y, regression=reg, autolag="AIC")
    print(f"\nADF test regression = {reg}")
    print("-----")
    print(f"ADF Statistic: {result[0]:.4f}")
    print(f"p-value: {result[1]:.4f}")
    print(f"Used Lag: {result[2]}")
    print("Critical Values:")
    for key, value in result[4].items():
        print(f"{key}: {value:.4f}")
```

```
ADF test regression = c
-----
ADF Statistic: -1.1101
p-value: 0.7110
Used Lag: 16
Critical Values:
1%: -3.4580
5%: -2.8737
10%: -2.5733
```

```
ADF test regression = ct
-----
ADF Statistic: -2.8951
p-value: 0.1638
Used Lag: 16
Critical Values:
1%: -3.9972
5%: -3.4290
10%: -3.1379
```